

Title

Patient comprehension and preferences for prostate cancer polygenic risk communication: findings from qualitative evaluation of prototype reporting materials

Abstract

As polygenic risk scores increasingly enter clinical prostate-cancer screening workflows, effective communication of complex genetic risk information becomes critical for informed decision-making. We conducted semi-structured qualitative interviews with eight men nationally representative of prostate-cancer-screening-eligible populations to evaluate comprehension, perceived usefulness, and visual-design preferences for prototype polygenic-risk summary materials and accompanying educational resources. Participants reviewed draft communication materials combining narrative text, numerical risk estimates, and graphical representations (pictograph arrays, probability curves, risk tables). Interviews explored interpretation accuracy, emotional responses, design-element preferences, and perceived actionability of the materials. Participants generally understood core risk-communication concepts and found simple graphical displays--particularly pictograph risk arrays--more intuitively interpretable than complex probability curves or bell-curve distributions. Color coding and visual hierarchy effectively directed attention to salient information. Several participants expressed ambiguity about recommended next steps despite clear written guidance. Participants valued supplementary resources such as embedded hyperlinks and decision-support frameworks. Design modifications addressing font size and information density were suggested. These pilot findings provide practical design guidance for communicating polygenic information as PRS-informed prostate-cancer screening moves toward clinical implementation.